

Electrical Characteristics

(V_{BATT} = 2.16V to 4.9V, typical value at 3.6V (Note 1), T_A = -40°C to +85°C, typical values are T_A = +25°C, see the *Functional Diagram*. Limits are 100% tested at T_A = +25°C. Limits over the operating temperature range and relevant supply voltage range are guaranteed by design and characterization.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
POWER SUPPLY						
Supply Voltage	V_{BATT}	(Note 1)	2.16		4.9	V
Undervoltage Shutdown Supply Current	I_{DD0}	Undervoltage shutdown		0.5	1.1	μ A
DeepShip Supply Current	I_{DD1}	$T_A \leq +50^\circ\text{C}$, typical at +25°C		0.5	1.1	μ A
Ship Supply Current	I_{DD2}	DpShpEn = 0, $T_A \leq +50^\circ\text{C}$, typical at +25°C, protection FETs off	1.4s updates	11	24	μ A
			5.625s updates	8		
Hibernate Supply Current	I_{DD3}	$T_A \leq +50^\circ\text{C}$, typical at +25°C, average current, CHG and DIS on, 1.4s updates		21	42	μ A
Active Supply Current	I_{DD4}	$T_A \leq +50^\circ\text{C}$, typical at +25°C, average current, not including thermistor measurement current		35	58	μ A
Regulation Voltage	V_{REG}			1.8		V
CHARGE ACCURACY						
Charge Voltage Accuracy	V_{GERR}	nVChgCfg1/2 setting, $T_A = +25^\circ\text{C}$	-7.5		+7.5	mV
		nVChgCfg1/2 setting, $-40^\circ\text{C} \leq T_A \leq +85^\circ\text{C}$	-20		+20	
Charge Voltage Range	V_{FS}	nVChgCfg1/2 setting, 5mV resolution	3.56		4.835	V
Charge Current Accuracy	I_{GERR}	nIChgCfg1/2 vs. CSP-CSN, Charge Current set 10mV to 25.6mV	-1.1		+1.1	%
		nIChgCfg1/2 vs. CSP-CSN, Charge Current set 6mV to 10mV	-1.15		+1.15	
	I_{GERR}	nIChgCfg1/2 vs. CSP-CSN, Charge Current from 4mV to 6mV	-1.25		+1.25	% of Reading
	I_{GERR}	nIChgCfg1/2 vs. CSP-CSN, Charge Current from 2.5mV to 4mV	-1.4		+1.4	%
Charge Current Range	I_{FS}	nIChgCfg1/2 setting, 400mA to 2560mA, 10mA steps (with 10m Ω)	2.5		25.6	mV
Charge Heat Regulation Max Setting		nChgCfg1.HeatLim; $R_{SENSE} = 10\text{m}\Omega$; multiply $R_{SENSE}/10\text{m}\Omega$ for other sense resistors		3264		mW
ANALOG-TO-DIGITAL CONVERSION						
Voltage Measurement Error	V_{GERR}	$T_A = +25^\circ\text{C}$, $2.3\text{V} \leq V_{BATT} \leq 4.9\text{V}$	-7.5		+7.5	mV
		$-40^\circ\text{C} \leq T_A \leq +85^\circ\text{C}$, $2.3\text{V} \leq V_{BATT} \leq 4.9\text{V}$	-20		+20	
Voltage Measurement Resolution	V_{LSB}			78.125		μ V
Current Measurement Offset Error	I_{OERR}	CSP = CSN = 4.4V, long-term average	-7.5	± 1.5	+7.5	μ V
Current Measurement Gain Error	I_{GERR}	CSP between CSN-50mV and CSN+50mV	-1		+1	% of Reading

Electrical Characteristics (continued)

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PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Current Measurement Resolution	I _{LSB}			1.5625			μV
Current Measurement Range	I _{FS}			±51.2			mV
Internal Temperature Measurement Error	T _{IGERR}			±1			°C
Internal Temperature Measurement Resolution	T _{ILSB}	TH (Note 2)		0.00391			°C
Auxiliary Ratiometric Measurement Error	T _{EGERR}			-0.5		+0.5	% of Reading
PCKP Measurement Resolution	V _{PLSB}			312.5			μV
PCKP Measurement Range	V _{PFS}			1.5		BATT + 5.12	V
PCKP Versus BATT Measurement Error	V _{P2Berr}	T _A = +25°C		-10		+10	mV
PCKP							
PCKP Startup Voltage				3.1			V
PCKP Startup Hysteresis				100	170	220	mV
PCKP Current		BATT = PCKP	T _A < +85°C, typical at T _A = +25°C	1.2		2.5	μA
PCKP Pulldown Resistor	R _{PD} PCKP			24	40	60	kΩ
CHARGE PUMP							
CP Output Voltage	V _{CP}	Battery only	I _{CHG} + I _{DIS} = 1μA	2 x V _{BATT} - 0.4	2 x V _{BATT} - 0.2	2 x V _{BATT}	V
CHG DRIVER							
CHG Output High	V _{OHC}	I _{OH} = -1mA		V _{CP} - 0.4			V
CHG Output Low	V _{OLC}	I _{OL} = 1mA		BATT + 0.4			V
DIS DRIVER							
DIS Output High	V _{OHD}	I _{OH} = -100μA		V _{CP} - 0.4			V
DIS Output Low	V _{OLD}	I _{OL} = 100μA		0.1			V
ZERO-VOLT CHARGE							
Voltage Drop Between ZVC and BATT	V _{ZVCDROP}	30mA into ZVC	BATT = 0V	1.4		2	V
			BATT = 2.3V	0.15		0.5	

Electrical Characteristics (continued)

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PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
INPUT/OUTPUT						
Output Drive Low, ALRT, SDA, PFAIL	V _{OL}	I _{OL} = 4mA, V _{BATT} = 2.3V	0.01		0.4	V
Output Drive High, PFAIL	V _{OH}	I _{OH} = -1mA, V _{BATT} = 2.3V	V _{BATT} - 0.1			V
Input Logic High, SCL, SDA, PIO	V _{IH}		1.5			V
Input Logic Low, SCL, SDA, PIO	V _{IL}				0.5	V
PIO Wake Debounce	PIO_WD	Ship mode		100		ms
External Thermistance Resistance	R _{EXT10}	nPackCfg.R100 = 0		10		kΩ
	R _{EXT100}	nPackCfg.R100 = 1		100		
RESISTANCE AND LEAKAGE						
Leakage Current, CSN, CSP, ALRT, TH	I _{LEAK}	V _{ALRT} < 15V	-1		+1	μA
Input Pulldown Current	I _{PD}	SDA, SCL pin = 0.4V		0.2	0.5	μA
COMPARATORS						
Overcharge Current Threshold Offset Error	OC _{OE}	OC comparator	-0.8		+0.8	mV
Overdischarge Current Threshold Offset Error	OD _{OE}	OD comparator	-1.5		+1.5	mV
Short-Circuit Threshold Offset Error	SC _{OE}	SC comparator	-2.5		+2.5	mV
Overcurrent Threshold Gain Error	ODOCSC _{GE}	OC, OD, or SC comparator	-4.0		+4.0	% of Threshold
Overcurrent Comparator Delay	OC _{DLY}	OD or SC comparator, 20mV minimum input overdrive, delay configured to minimum		2	6	μs
Supplement Mode Comparator Threshold Falling PCKP Versus BATT	V _{SUP_TH_F}	BATT ≥ 3.4V, PCKP sweep down		30		mV
TIMING						
Time-Base Accuracy	t _{ERR}	T _A = +25°C	-1		+1	%
SHA Calculation Time	t _{SHA}	V _{BATT} > 3V		4.5	10	ms
TH Precharge Time	t _{PRE}	Time between turning on the TH bias and analog-to-digital conversions	8.48			ms
Task Period	t _{TP}			351.5		ms
NONVOLATILE MEMORY						
Nonvolatile Access Voltage	V _{NVM}	For block programming and recalling, applied on BATT	3.0			V
Programming Supply Current	I _{PROG}	Current from BATT at 2.9V for block programming	2	5.5	10	mA

Electrical Characteristics (continued)

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PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Block Programming Time	t_{BLOCK}			368	7360	ms
Page Programming Time	t_{UPDATE}	SHA secret update or learned parameters update		64	1280	ms
Nonvolatile Memory Recall Time	t_{RECALL}				5	ms
Write Capacity, Configuration Memory	n_{CONFIG}	(Notes 2, 3, 4)		7		writes
Write Capacity, SHA Secret	n_{SECRET}	(Notes 2, 3, 4)		5		writes
Write Capacity, Learned Parameters	$n_{LEARNED}$	(Notes 2, 3, 4)		99		writes
Data Retention	t_{NV}	(Note 2)	10			years
1-WIRE INTERFACE, REGULAR SPEED						
Time Slot	t_{SLOT_STD}		60		120	μ s
Recovery Time	t_{REC_STD}		1			μ s
Write-0 Low Time	t_{LOW0_STD}		60		120	μ s
Write-1 Low Time	t_{LOW1_STD}		1		15	μ s
Read-Data Valid	t_{RDV_STD}				15	μ s
Reset-Time High	t_{RSTH_STD}		480			μ s
Reset-Time Low	t_{RSTL_STD}		480		960	μ s
Presence-Detect High	t_{PDH_STD}		15		60	μ s
Presence-Detect Low	t_{PDL_STD}		60		240	μ s
1-WIRE INTERFACE, OVERDRIVE SPEED						
Time Slot	t_{SLOT_OVD}		6		16	μ s
Recovery Time	t_{REC_OVD}		1			μ s
Write-0 Low Time	t_{LOW0_OVD}		6		16	μ s
Write-1 Low Time	t_{LOW1_OVD}		1		2	μ s
Read-Data Valid	t_{RDV_OVD}				2	μ s
Reset-Time High	t_{RSTH_OVD}		48			μ s
Reset-Time Low	t_{RSTL_OVD}		48		80	μ s
Presence-Detect High	t_{PDH_OVD}		2		6	μ s
Presence-Detect Low	t_{PDL_OVD}		8		24	μ s
2-WIRE INTERFACE						
SCL Clock Frequency	f_{SCL}	(Note 5)	0		400	kHz
Bus Free Time Between a STOP and START Condition	t_{BUF}		1.3			μ s
Hold Time (Repeated) START Condition	$t_{HD:STA}$	(Note 6)	0.6			μ s

Electrical Characteristics (continued)

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PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Low Period of SCL Clock	t_{LOW}		1.3			μ s
High Period of SCL Clock	t_{HIGH}		0.6			μ s
Setup Time for a Repeated START Condition	$t_{SU:STA}$		0.6			μ s
Data Hold Time	$t_{HD:DAT}$	(Notes 7, 8)	0		0.9	μ s
Data Setup Time	$t_{SU:DAT}$	(Note 7)	100			ns
Rise Time of Both SDA and SCL Signals	t_R		5		300	ns
Fall Time of Both SDA and SCL Signals	t_F		5		300	ns
Setup Time for STOP Condition	$t_{SU:STO}$		0.6			μ s
Spike Pulse Width Suppressed by Input Filter	t_{SP}	(Note 9)			50	ns
Capacitive Load for Each Bus Line	C_B				400	pF
SCL, SDA Input Capacitance	C_{BIN}			6		pF

Note 1: All voltages are referenced to GND.

Note 2: Specification is guaranteed by design (GBD) and not production tested.

Note 3: Write capacity numbers shown have one write subtracted for the initial write performed during manufacturing test to set nonvolatile memory to a known value.

Note 4: Due to the nature of one-time programmable memory, write endurance cannot be production tested. Follow the nonvolatile memory and SHA secret update procedures detailed in the data sheet.

Note 5: Timing must be fast enough to prevent the IC from entering shutdown mode due to bus low for a period greater than the shutdown timer setting.

Note 6: f_{SCL} must meet the minimum clock low time plus the rise/fall times.

Note 7: The maximum $t_{HD:DAT}$ has to only be met if the device does not stretch the low period (t_{LOW}) of the SCL signal.

Note 8: This device internally provides a hold time of at least 100ns for the SDA signal (referred to the minimum V_{IH} of the SCL signal) to bridge the undefined region of the falling edge of SCL.

Note 9: On 10m Ω R_{SENSE} .